JNTUH USED 21-05-18 AM Set 2

R13

Code No: 117BX

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, April/May - 2018
COMPUTER GRAPHICS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) How are computer graphics useful in image processing and Visualization? [2]
b) Explain two successive translation vectors are additive. [3]
c) Write any two advantages of cohen sutherland algorithm. [2]
d) Write about Homogeneous coordinates. [3]
e) Define B spline surface. [2]
f) Give the properties of B-Spline curves. [3]
g) What are the advantages of BSP Tree methods? [2]
h) Write about area sub division method. [3]
i) Define Computer Animation. [2]
j) Write about any two computer animation languages. [3]

PART-B

(50 Marks)

2.a) Describe the construction and working principle of CRT Raster Scan system with a neat sketch.
b) What are the steps involved in depth buffer algorithm? What are its advantages and disadvantages? [5+5]

OR

3.a) Explain about the Bressen hams line generation algorithm.
b) Describe about the flat panel display devices. [5+5]

4.a) Clip a line segment between points (1, 3) to (5, 17) using Cohen Sutherland clipping algorithm so that it fit into view port with left bottom at (2, 5) and right top at (5, 12).
b) What are the steps involved in reflecting the object about an arbitrary axis using 2-D transformations? [5+5]

OR

5.a) Show that a rotation about the origin can be done by performing three shearing Transformations
b) Derive the window to view port transformations equations by first scaling the window to the size of the viewport and then translating the scaled window to the view port position. [5+5]

6.a) Determine the Bezier blending functions for five control points. Plot each function and label the maximum and minimum values.
b) Compare the characteristics of Beizer and B-spline curves. [5+5]

OR

7.a) Differentiate between Bezier curve and B-Spline curve.
b) Explain about Hermite curve with equations and examples. [5+5]

8.a) Explain how the issues involved in rotation and reflection are different in 3-D from 2-D. Give a detailed note.
b) Derive the transformation matrix for perspective transformation. [5+5]

OR

9.a) Explain in brief about octree methods.
b) Write a program that uses the depth-sorting method to display the visible surfaces of any given object with plane faces. [5+5]

10.a) Define the term morphing and explain its use in key frame systems of Animation.
b) Briefly explain the steps involved in the design of animation sequence. [5+5]

OR

11.a) Explain in brief about computer animation functions.
b) What are the various types of interpolation used in animation? [5+5]

--ooOoo--

JNTUH USED 21-05-18AM Set 2